

CS224N: Natural Language Processing with Deep Learning

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Lecture 1 Notes - Introduction to Natural Language Processing & Word Vectors

Unit 1: Introduction to Natural Language Processing & Semantic Representations

Natural language processing (NLP) is an interdisciplinary subfield of computer science, artificial intelligence, and

Historically, early NLP systems relied on symbolic rules and discrete representation of words as atomic symbols.

Unit 2: Limitations of One-Hot Vectors & The Orthogonality Problem

While one-hot representation provides a straightforward structural encoding for words, it exhibits severe fundamental limitations. In a one-hot vector space, every word vector is orthogonal to every other word vector. Consequently, the dot product of any two word vectors is zero. Because orthogonal vectors share no inner-product similarity, one-hot encodings cannot represent word similarity.

Unit 3: Distributional Semantics & Firth's Distributional Hypothesis

To address the orthogonality limitation of discrete representations, modern NLP relies on distributional semantics.

Distributional semantics is founded on J.R. Firth's distributional hypothesis (1957): 'You shall know a word by the

Unit 4: Word2Vec Framework Overview & Objective Function

Word2Vec is a computationally efficient framework introduced by Mikolov et al. (2013) at Google for learning continuous word representations. The central principle of Word2Vec is to iterate through a text corpus using a sliding context window of size c . For

Unit 5: Continuous Bag-of-Words (CBOW) Architecture

The Word2Vec framework defines two main model architectures: Continuous Bag-of-Words (CBOW) and Contin

In the Continuous Bag-of-Words (CBOW) model, the network predicts a target center word given a set of surround

Unit 6: Continuous Skip-Gram Architecture

The Continuous Skip-gram architecture operates inversely to CBOW: given a single center word w_t , the model For each position t in a corpus, Skip-gram computes the conditional probability $P(w_{t+j} | w_t)$ of context words

Unit 7: Computational Bottlenecks & Softmax Optimization

Calculating the exact Softmax conditional probability denominator $\sum_{w=1}^{|V|} \exp(u_w^T v_c)$ in standard S

Because vocabulary size $|V|$ routinely exceeds several hundred thousand or millions of words, computing the full

Unit 8: Negative Sampling (NEG) Formulation & Objective

To solve the Softmax computational bottleneck, Mikolov et al. proposed Negative Sampling (NEG), derived from
Negative Sampling simplifies the multi-class classification problem into a set of binary logistic regression tasks. F

Unit 9: Co-Occurrence Matrices & Global Count-Based Methods

In contrast to local context window prediction models like Word2Vec, traditional statistical NLP relied on global co-

A term-context co-occurrence matrix X is a large matrix where entry X_{ij} records the number of times word i appears in context j .

Unit 10: Pointwise Mutual Information (PMI) & PPMI Transformation

To normalize raw co-occurrence frequencies against chance word associations, statistical models calculate Pointwise Mutual Information (PMI). The PMI between word w and context c is defined as $\log(P(w,c) / (P(w)P(c)))$. PMI measures how much more frequent the pair (w,c) is than expected by chance.

Unit 11: GloVe Architecture & Log-Bilinear Model

The Global Vectors for Word Representation (GloVe) model introduced by Pennington, Socher, and Manning (2017).

GloVe trains a log-bilinear model directly on non-zero entries of global co-occurrence matrix X . The objective function is:

Unit 12: Linear Vector Arithmetic & Word Analogy Benchmarks

A remarkable emergent property of continuous word vector spaces trained on large corpora is linear vector arithmetic. Dense word embeddings preserve semantic and syntactic relations as constant vector offsets in d -dimensional space.

Unit 13: Intrinsic vs Extrinsic Word Vector Evaluation

Evaluating word vector quality is conducted using two distinct benchmark paradigms: intrinsic evaluation and extrinsic evaluation.

Intrinsic evaluation tests word vector quality directly on isolated subtasks without integrating vectors into downstream tasks.

Unit 14: Subword Embeddings & FastText Architecture

Standard word-level embedding models assign a single vector to each word in vocabulary V , failing to handle out-of-vocabulary words. To solve the out-of-vocabulary problem, FastText (Bojanowski et al., 2017) extends Skip-gram by representing each word as a vector of subword embeddings.